

PREFACE

This 64th Edition of the IATA *Dangerous Goods Regulations* becomes effective on 1 January 2023 and replaces the 63rd Edition, which must not be used after 31 December 2022 unless specifically permitted in these Regulations.

The IATA *Dangerous Goods Regulations* are published by the IATA Dangerous Goods Board pursuant to IATA Resolutions 618 and 619 and constitute a manual of industry carrier regulations to be followed by all IATA Member airlines. This edition of the IATA Regulations is based on the requirements of Annex 18 to the Convention on International Civil Aviation (Chicago, 1944) and the 2023—2024 Edition of the associated *Technical Instructions for the Safe Transport of Dangerous Goods by Air* (Doc 9284).

Annex 18 to the Chicago Convention and the associated *Technical Instructions for the Safe Transport of Dangerous Goods by Air* are recognized as the sole authentic legal source material in the air transport of dangerous goods. Consequently, any additional or explanatory material added by IATA does not form part of the authentic text of the ICAO *Technical Instructions* and does not have the same legal force.

In developing its Regulations, IATA has drawn on its extensive experience to give special attention to the format and wording of these Regulations to make this a readily understandable and easy-to-use manual. There are certain differences between the IATA and ICAO regulations which stem from operational considerations and result in a regulatory regime which is necessarily more restrictive than the ICAO requirements. These differences are identified by the symbol  appearing in the margin. The IATA Regulations also incorporate additional material of practical assistance to users.

The IATA *Dangerous Goods Regulations* are also available in Chinese, French, German, Russian and Spanish language versions. A Japanese language edition is also produced under licence by the Japan Air Cargo Institute for Safety Inc. (JACIS).

Any comments concerning this Manual should be addressed to:

The Editor, Dangerous Goods Regulations
International Air Transport Association
Cargo Department
Route de l'Aéroport 33
PO Box 416
1215 Geneva Airport
SWITZERLAND
Tel: +1 (514) 390 6726
E-mail: dangood@iata.org

For Internet dangerous goods information (web address):

www.iata.org/dangerousgoods

Dangerous goods support line:

Tel: +1 (514) 390 6726
E-mail: dangood@iata.org



ACKNOWLEDGEMENTS

These IATA *Dangerous Goods Regulations* have been developed by the IATA Dangerous Goods Board pursuant to authority of the IATA Cargo Services Conference. The composition of this Board is:

Mr. E. Boon, KLM Royal Dutch Airlines, Amsterdam, Netherlands
 Ms. C. Costa Diniz, DHL Express, Brussels, Belgium
 Mr. P. Ehrensperger, Swiss International Airlines, Zurich, Switzerland
 Ms. T. Farquharson, Air New Zealand, Auckland, New Zealand
 Mr. T. Howard, Emirates Airline, Dubai, United Arab Emirates
 Mr. N. Iki, Japan Airlines, Tokyo, Japan
 Ms. F. Lam, Cathay Pacific Airways, Hong Kong, China SAR
 Mr. T. Leech, FedEx Express, Memphis, USA
 Mr. A. McCulloch, UPS Airlines, Louisville, USA
 Mr. G. Sokayan, Singapore Airlines, Singapore
 Mr. M. Tobin, Alaska Airlines, Seattle, USA

IATA Secretariat:

Mr. D. Brennan, Head, Cargo Safety & Dangerous Goods
 Ms. C. Chan, Manager, Dangerous Goods Standards
 Ms. P. Nielsen, Manager, Dangerous Goods & Cargo Training Standards

It should be noted that these IATA *Dangerous Goods Regulations* are subject to constant review in the light of new technical developments and changing requirements of industry and air transportation.

Record of Addendum

From time-to-time there is a need to issue an addendum to the *Dangerous Goods Regulations* (DGR) to incorporate changes issued by ICAO to the contents of the *Technical Instructions for the Safe Transport of Dangerous Goods by Air* or to reflect changes to the content of the DGR. A record of addendum table is provided below to assist holders of the DGR to record that the addendum has been received and incorporated.

RECORD OF ADDENDUM

Addendum No.	Date ^a	Name ^b	Addendum No.	Date	Name

^a The date the addendum changes were included into this copy of the DGR.

^b The name of the person who included the addendum changes into this copy of the DGR.

Note:

Because the Digital DGR will update automatically when amendments (addendum) and bug fixes are released, the date for each addendum will be left blank. The publication will only be refreshed with the updates when there is an Internet connection.



INTRODUCTION

PURPOSE OF THE DANGEROUS GOODS REGULATIONS

The IATA *Dangerous Goods Regulations* are published in order to provide procedures for the shipper and the operator by which articles and substances with hazardous properties can be safely transported by air on all commercial air transport.

In 1953, the Member airlines of IATA recognized the growing need to transport by air, articles and substances having hazardous properties which, if uncontrolled, could adversely affect the safety of the passengers, crew and/or aircraft on which they are carried. Experience in other modes of transport had demonstrated that most such articles and substances could be carried safely provided that the article or substance was properly packed and the quantities in each package were properly limited. Using this experience together with the industry's knowledge of the specialized characteristics of air transport, IATA developed the first regulations for the transport of dangerous goods by air. The first edition of the IATA *Dangerous Goods Regulations* was published in 1956 as the IATA Restricted Articles Regulations.

GENERAL PHILOSOPHY

Dangerous goods can be transported safely by air transport provided certain principles are strictly followed. The IATA *Dangerous Goods Regulations* is an easy-to-use manual based on the International Civil Aviation Organization (ICAO) *Technical Instructions for the Safe Transport of Dangerous Goods by Air*. It incorporates additional operational requirements, which provide a harmonized system for operators to accept and transport dangerous goods safely and efficiently.

The Regulations include a detailed list of individual articles and substances specifying the United Nations classification of each article or substance and their acceptability for air transport as well as the conditions for their transport. Since no listing can be complete, the list also includes many generic or "not otherwise specified" entries to assist in the classification of those articles or substances not listed by name.

Some dangerous goods have been identified as being too dangerous to be carried on any aircraft under any circumstances; others are forbidden under normal circumstances but may be carried with specific approvals or exemptions from the States concerned; some are restricted to carriage on all cargo aircraft; most however, can be safely carried on passenger aircraft as well, provided certain requirements are met.

Packaging is the essential component in the safe transport of dangerous goods by air. The IATA *Dangerous Goods Regulations* provide packing instructions for all dangerous goods acceptable for air transport with a wide range of options for inner, outer and single packagings. The packing instructions normally require the use of UN performance-tested specification packagings, however these are not required when dangerous goods are shipped in Limited Quantities under the provisions of Limited Quantity "Y" Packing Instructions. The quantity of dangerous goods permitted within these packagings is strictly limited by the Regulations so as to minimize the risk should an incident occur.

Training is also an essential element in maintaining a safe regulatory regime. It is necessary for all individuals involved in the preparation or transport of dangerous goods to be trained in the functions for which they are responsible. Depending on the job-function, this may entail only familiarization training or may also include more detailed training in the intricacies of the Regulations. It is important to remember that dangerous goods are very unlikely to cause a problem when they are prepared and handled in compliance with the IATA *Dangerous Goods Regulations*.

The proper declaration of dangerous goods by the shipper ensures that all in the transportation chain know what dangerous goods they are transporting, how to properly load and handle them and what to do if an incident or accident occurs either in-flight or on the ground. The pilot-in-command must know what is on board the aircraft in order to properly deal with any emergencies, which may occur. The pilot must also convey this information, if possible, to air traffic services to aid in the response to any aircraft incident or accident.



Dangerous Goods Regulations

Information regarding dangerous goods forbidden in baggage must also be conveyed to passengers to assist them in recognizing dangerous goods, which they are not permitted to carry on their person or in their baggage and which may not be readily recognizable as being dangerous.

Lastly, dangerous goods accidents or incidents must be reported, so that an investigation by the relevant authorities can establish the cause and take corrective action. Also, if as a result of these investigations changes are required in the Regulations, appropriate regulatory action can be taken without delay.

SIGNIFICANT CHANGES AND AMENDMENTS IN THE 64TH EDITION (2023)

The 64th edition of the IATA *Dangerous Goods Regulations* incorporates all amendments made by the ICAO Dangerous Goods Panel in developing the content of the 2023–2024 edition of the ICAO Technical Instructions as well as changes adopted by the IATA Dangerous Goods Board. The following list is intended to assist the user to identify the main changes introduced in this edition and must not be considered an exhaustive listing. The changes have been prefaced by the section or subsection in which the change occurs.

Record of Addendum. Notes have been added to the record of addendum table to clarify that the date entered in the table should be the date that the addendum changes were incorporated into the copy of the DGR and the name entered is that of the person who incorporated those changes.

2—Limitations

2.3—Dangerous Goods Carried by Passengers or Crew—The provisions for battery-powered mobility aids (2.3.2.2–2.3.2.4) have been revised to identify that batteries only need be removed when the design of the mobility aid does not provide adequate protection against damage to the battery.

Note:

The requirement for mobility aids with spillable batteries (2.3.2.3) to be handled, loaded and stowed upright at all times, or the battery must be removed remains.

2.6—Dangerous Goods in Excepted Quantities—Paragraph 2.6.5.1 has been revised to clarify that a package containing dangerous goods in excepted quantities may also contain goods not subject to the Regulations.

2.8.1—State Variations—The List (2.8.1) and List of State Variations (2.8.2) have been revised to include new States and amended variations for existing States.

3—Classification

3.4.1.2.4 and 3.5.2.2—The classification provisions for self-reactive substances and organic peroxides, respectively, have been revised to fully align to the wording in the UN Model Regulations.

3.8.3.2.3—This paragraph has been revised to clarify that substances or mixtures classified as corrosive must be assigned to Packing Group I if the test results do not indicate a different packing group.

3.9.2.6.1—The requirements of the lithium battery test summary have been revised to remove the need for the test summary for button cells installed in equipment.

4—Identification

4.2—List of Dangerous Goods

The amendments to the List of Dangerous Goods include:

- change of the maximum net quantity per package on Cargo Aircraft Only for UN 2794, **Batteries, wet, filled with acid**, UN 2795, **Batteries, wet, filled with alkali** and UN 3292, **Cells, containing sodium** from “No limit” to 400 kg. These articles must be packed in a UN specification packaging, for which the limit is 400 kg.
- addition of Special Provision A154 to the entries for UN 3171, **Battery-powered vehicle**, UN 3528, UN 3529, UN 3530, engines and machinery, UN 2990 and UN 3072, life-saving appliances and UN 3166, vehicles.
- addition of a new entry, UN 3550, **Cobalt dihydroxide powder**.
- addition of Special Provision A4 to UN 2922, **Corrosive liquid, toxic, n.o.s.** and addition of A5 to UN 2923, **Corrosive solid, toxic, n.o.s.**
- reclassification of UN 1891, **Ethyl bromide**, from Division 6.1 to Class 3 with a Division 6.1 subsidiary hazard.
- deletion of UN 1169, **Extracts, aromatic, liquid** and change of the proper shipping name for UN 1197, to become **Extracts, liquid**, for flavour or aroma.
- addition of packing group(s), where applicable, to entries shown as “forbidden/forbidden” with a UN number and proper shipping name. The packing group, when assigned, forms part of the classification information.

4.4—Special Provisions

The amendments to the special provisions include:

- addition of a sentence to special provisions A1 and A2 specifying that when dangerous goods are being offered under an approval that A1 or A2, as applicable must be noted in the authorization column of the Shipper's Declaration.
- amendment to A176 specifying that when metal hydride storage systems are being offered that the special provision number must be noted in the authorization column of the Shipper's Declaration;

New special provisions include:

- A221—assigned to UN 1002, **Air, compressed** identifying that mixtures of nitrogen and oxygen in the concentrations specified may be transported as UN 1002 and that a Division 5.1 subsidiary hazard label is not required;
- A223—assigned to UN 2990, **Life-saving appliance, self-inflating** and UN 3072, **Life-saving appliance, not self-inflating** identifying that life-saving appliances packed in strong rigid outer packagings with no dangerous goods other than cartridges of a Division 2.2 compressed or liquefied gas, with no subsidiary hazard installed for the purposes of the activation of the device may be shipped as cargo as "not restricted";

Note:

the wording of A223 was previously included in Packing Instruction 955.

- A224—assigned to UN 3548, **Articles containing miscellaneous dangerous goods, n.o.s.** This new special provision identifies that articles containing an environmentally hazardous substance may be shipped as cargo on a passenger or cargo aircraft in accordance with the provisions of Packing Instruction 975 and that Special Provision A2 assigned to the UN number then does not apply;
- A225—assigned to UN 3538, **Articles containing non-flammable, non-toxic gas, n.o.s.** This new special provision identifies that articles containing a Division 2.2 gas without a subsidiary hazard may be shipped as cargo on a passenger or cargo aircraft in accordance with the provisions of Packing Instruction 222 and that Special Provision A2 assigned to the UN number then does not apply.

5—Packing

Packing Instructions

Table 5.0.B—List of Inner Packagings—Has been amended to remove reference to metal and plastic aerosols (IP7, IP7A, IP7B and IP7C).

PI 203, PI Y203 and PI Y963—Have been revised to remove all details of metal and plastic aerosols (IP7, IP7A, IP7B and IP7C). There is now a standard capacity of 1,000 mL for metal aerosols and receptacles, small containing gas (gas cartridges) and 500 mL for plastic receptacles in line with the provisions of the UN Model Regulations.

PI 220, PI 378 and PI 972—Have been revised to include additional packing requirements identifying that unless the design and construction of the engine or machinery is such that the containment for the dangerous goods is adequately protected then the engine or machinery must be packed in strong outer packagings or fixed in cradles, crates or other handling devices. Additional provisions have been added such that engines or machinery are forbidden for transport if lithium batteries in the engine or machinery are damaged or defective. There is also now a specific reference to engines or machinery that contain pre-production prototype lithium cells or batteries or low production runs of lithium cells or batteries that may be shipped on a cargo aircraft under an approval from the appropriate authority of the State of origin and the State of the operator.

PI 222 and PI 975—Have been added to address the packing provisions for UN 3538, **Articles containing non-flammable, non-toxic gas, n.o.s.** and UN 3548, **Articles containing miscellaneous dangerous goods, n.o.s.**, respectively, provided that the articles meet the conditions specified in Special Provision A224 or A225.

PI 870—Has been revised to remove reference to the packing provisions for batteries installed in equipment. Where batteries are installed in equipment, the correct classification is UN 3171, **Battery-powered equipment** and PI 952 applies.

PI 950, PI 951 and PI 952—Have been revised identifying that vehicles are forbidden for transport if lithium batteries in the vehicle are damaged or defective. There is also now a specific reference for vehicles that contain pre-production prototype lithium cells or batteries or low production runs of lithium cells or batteries

that may be shipped on a cargo aircraft under an approval from the appropriate authority of the State of origin and the State of the operator.

PI 965 and PI 968—Have been revised to require that each package prepared in accordance with Section IB must be capable of withstanding the 3 m stack test as applies to limited quantity packages.

PI 966, PI 967, PI 969 and PI 970—Section II have been revised to specify that when packages are placed into an overpack that the packages must be secured in the overpack and the intended function of each package must not be impaired by the overpack. This aligns to the general requirements for overpacks specified in 5.0.1.5.

6—Packaging Specifications and Performance Tests

6.1.7—6.1.9—The requirements for aerosols have been deleted.

6.4.1, 6.4.2—The provisions for the design, construction, testing, initial and periodic inspection of cylinders, including UN cylinders have been revised.

6.4.4—The provisions for the test requirements for aerosols and gas cartridges have been revised.

7—Marking and Labelling

7.1.5.5—The lithium battery mark has been revised to remove the requirement for a telephone number to be provided on the mark. There is a transition period until 31 December 2026 during which time the mark shown in the 63rd edition of the DGR may continue to be used.

8—Documentation

8.2.1—The transitional period for the dangerous goods statement on the air waybill has been extended by a further two years until 31 December 2024.

9—Handling

9.3.7—The provisions for replacements of labels on packages where the labels are found to be lost, detached or illegible after acceptance have been revised to include a requirement for marks, such as the excepted quantity mark and limited quantity mark to be replaced by the operator.

10—Radioactive Materials

10.8.3.9.2, Step 6—The wording of Step 6 has been revised to include reference to mixtures of radionuclides for which relevant data are not available and where the shipper has determined the A_1 or A_2 value using Table 10.3.B that a statement must be added to the Shipper's Declaration.

10.8.3.9.4—Step 13 has been revised to include reference to mixtures of radionuclides for which relevant data are not available and where the shipper has used Table 10.3.B to determine the appropriate A_1 or A_2 value that reference must be made to the use of table 10.3.B as well as an indication of the radioactive contents as specified in the first column of table 10.3.B.

Appendix A—There are many changes to the defined terms shown in the Glossary.

Appendix C—There are additions to the list of currently assigned self-reactive substances on Division 4.1 in Table C.1 and to the list of organic peroxides in Table C.2.

Appendix D—Contact details for competent authorities have been updated.

Appendix E—Changes have been made to the list of UN Specification Packaging Suppliers (E.1) and the Package Testing Facilities (E.2).

Appendix F—The list of Sales Agents (F.2) has been revised. IATA Accredited Training Schools (F.3—F.5) have been updated to include entities that have joined the new CBTA Centre Program.

Appendix H—The guidance material on development and implementation of competency-based training for dangerous goods has been removed from the DGR and is now posted on the IATA website as a stand-alone document, www.iata.org/dangerousgoods.

REFERENCE MARKS

The following symbols placed against an item indicate changes from the previous edition:

Symbol—Meaning

□—Addition of a new item.

△—Change to an item.

⊗—Cancellation of an item.

✎—Additional IATA requirements.

☼—Indicates that the item relates entirely to Radioactive shipments.

ACCEPTANCE CHECKLIST

Paragraph 9.1.3 requires that operators use a checklist when accepting dangerous goods. The content of the checklist is the responsibility of each operator.

As a guide, dangerous goods acceptance checklists for radioactive and for non-radioactive shipments have been included at the end of these Regulations. In addition, a simplified checklist for dry ice has been included.

RELATIONSHIP OF THE SIX (6) LANGUAGE EDITIONS

The IATA *Dangerous Goods Regulations* are published in six languages: English, Chinese, French, German, Russian and Spanish. A Japanese language edition is also produced under licence by the Japan Air Cargo Institute for Safety Inc. (JACIS). The electronic version of the IATA *Dangerous Goods Regulations* is also available in English, French, German and Spanish.

The English text is used for the translations, therefore, if there is a difference between the English text and any of the other language texts, the English text prevails.

HOW TO USE THE REGULATIONS

The detailed content of these Regulations gives all the necessary provisions to enable a shipper to correctly prepare a consignment of dangerous goods for air transport and for a freight forwarder and operator/ground handling agent dangerous goods acceptance personnel to be able to verify, to the extent possible, that the consignment of dangerous goods complies with the requirements of these Regulations. To assist shippers the following step-by-step procedure is given for guidance to ensure all the applicable requirements for classifying, packing, marking, labelling and documenting are met.

The information given below is for guidance only and the relevant sections should be checked to ascertain their relevance to each consignment.

1. Determine the correct technical name or composition of the substance or the description of the article and check if substance is forbidden on aircraft (see Subsections 2.1 and 4.2).
2. Ascertain whether the name or composition of the article or substance appears in the List of Dangerous Goods in Subsection 4.2 and if so, its proper shipping name. That proper shipping name must then be used on the Shipper's Declaration and be marked on the package(s).
3. If only the UN number of the substance or article is known, Subsection 4.3 provides a cross-reference from the UN/ID number to the proper shipping name and is arranged in numerical order. For the language editions, Subsection 4.3 shows the proper shipping name applicable to the UN number, the proper shipping name in English, which must be used on the Shipper's Declaration for Dangerous Goods and the page number on which the entry appears in Table 4.2.
4. If the substance or article does not appear in the list, determine the class or division into which it falls by comparing its known properties with the definitions for the various classes, which are given in Section 3.
5. If the properties are not known, tests should be carried out to determine the appropriate class and division. If the article or substance is not listed by name in the list and does not meet the definition of any of the classes, it is not subject to these Regulations.
6. For articles or substances with multiple hazards, the provisions of Subsection 3.10 should be followed. Once all the properties of the article or substance are known, determine whether it is forbidden for transport according to the provisions of Subsections 2.1 and 4.2.
7. If the article or substance does not come within the provisions of Subsection 2.1, determine the proper shipping name from the most appropriate of the generic or n.o.s. entries in Table 4.1.A, in accordance with the hierarchy shown in Subsection 4.1. If the substance or article does not appear in the list, refer to Appendix C.1 and C.2 for further possible identification.
8. If the quantity of material to be transported is very small, it may be possible to utilise the provisions for Dangerous Goods in Excepted Quantities provided that all of the requirements of Subsection 2.6 are met.
9. Determine whether it is desired to transport the article or substance on passenger or cargo aircraft.
10. From the information given in Columns G to L of the List of Dangerous Goods, ascertain whether or not the article or substance is forbidden for transport on passenger aircraft or on both passenger and cargo aircraft.
11. For classification of Radioactive Material (Class 7) and the packaging, marking, labelling, certification and documentary requirements for radioactive shipments, consult Section 10.
12. If the article or substance is forbidden for transport on passenger aircraft, determine whether it can be transported on cargo aircraft.
13. For shipments intended for carriage on PASSENGER AIRCRAFT:
Having determined that the article or substance is not forbidden for carriage on passenger aircraft, determine the following:
 - Packing Instruction number (see Subsection 4.2, Column G and I);
 - Quantity limitation (see Subsection 4.2, Column H and J for package limits and the Packing Instruction for inner packaging limits);
 - Applicable State or operator Variations (see Subsection 2.8).
14. For shipments intended for carriage on CARGO AIRCRAFT (or which can only be carried on such aircraft):
Having determined that the article or substance is not forbidden for carriage on cargo aircraft, determine the following:
 - Packing Instruction number (see Subsection 4.2, Column K);
 - Quantity limitation (see Subsection 4.2, Column L for package limits and the Packaging Instruction for inner packaging limits);
 - Applicable State or operator Variations (see Subsection 2.8).

15. Determine the other packing details from the relevant information or packing instruction in Section 5 and any special requirements from Sections 1 and 4.

It should be noted that the Limited Quantity provisions may be used as an acceptable alternative to UN packagings, subject to State and Operator Variations. (See Subsection 2.7, 5.0.3, Subsections 6.1, 6.2 and 6.6).

16. Select, where permitted, a method of packing from the packing instruction, or ascertain the provisions of the instruction and ensure the packagings to be used meet all the relevant requirements of Subsection 5.0 and Section 6.
17. Note that State and operator Variations listed in the packing instructions do not necessarily constitute a complete list of applicable variations. Subsection 2.8 must always be checked to determine all applicable State and operator Variations.
18. Ensure all the appropriate markings and labels are affixed to or printed on the packages according to Section 7.
19. Complete and sign the Shipper's Declaration for Dangerous Goods in accordance with Section 8 and prepare the Air Waybill.
20. Offer the complete consignment for transport by air.





Dangerous Goods Regulations



Dangerous Goods Regulations